Backbone Detectives · Teacher and Cover Plan

Learning goal: I can use inside body clues to sort tricky animals and explain my choice.

Scheme Outcome

BASE Science scheme of work, Unit 1 (Skeletons, Muscles and Nutrition), Topic: Compare animals with and without a backbone (vertebrates and invertebrates).

Use structural body evidence to classify familiar and unfamiliar animals as vertebrates or invertebrates.

Success Criteria

I can check inside clues instead of guessing from legs or wings.

I can sort eight animals into backbone and no backbone.

I can explain my choice using because.

Model Sequence (I do)

Look at the snake x-ray. It has no legs, yet a long spine runs down its back.

That inside bone row proves the snake is a vertebrate.

Look at the lobster. It is large with legs, but its hard shell is outside with no spine inside.

A bat and a butterfly both have wings. Only the bat has bones and a spine inside.

If a picture hides the inside, say: 'I need an inside

clue to be sure.'

Use the proof stem: 'The __ is a __ because its clue shows __.'

We Do & Hinge Check

Is a snake a vertebrate? Choose the best reason.

Answer: A. A snake is a vertebrate because it has a backbone inside.

Safe Practical & Card Setup

Sort eight detective cards into two groups. Pick two tricky animals and prove your choices.

Kit: Eight creature detective cards with inside clues, two group mats, and sentence strips.

Keep cards on the table. Move calmly between desk stations.

You Do Routes (Supported, Independent, Aspire)

Supported: Sort six creature cards using the fish and crab anchor cards. Point to the clue that proves one choice. (Help: Does it match the fish (backbone inside) or the crab (no backbone)?)

Independent: Sort all eight creature cards. Pick two tricky animals and write a because sentence for each. (Help: The __ is a __ because the inside clue shows __.)

Aspire: Explain why a bat and a beetle are in different groups even though both fly. (Help: Both have wings, but inside a bat has __ while a beetle has

__.)